Exp No 6
11/3/26

FACTORIAL OF A NUMBER

AIM:

To write and execute an assembly language program to perform the factorial of a number using 8051 keil

APPARATUS REQUIRED:

Personal Computer with keil software

Algorithm:

(i) Start

(ii) Input : Read the number n

(iii) Initialize:

* Set factorial to 1
* Set i to 1

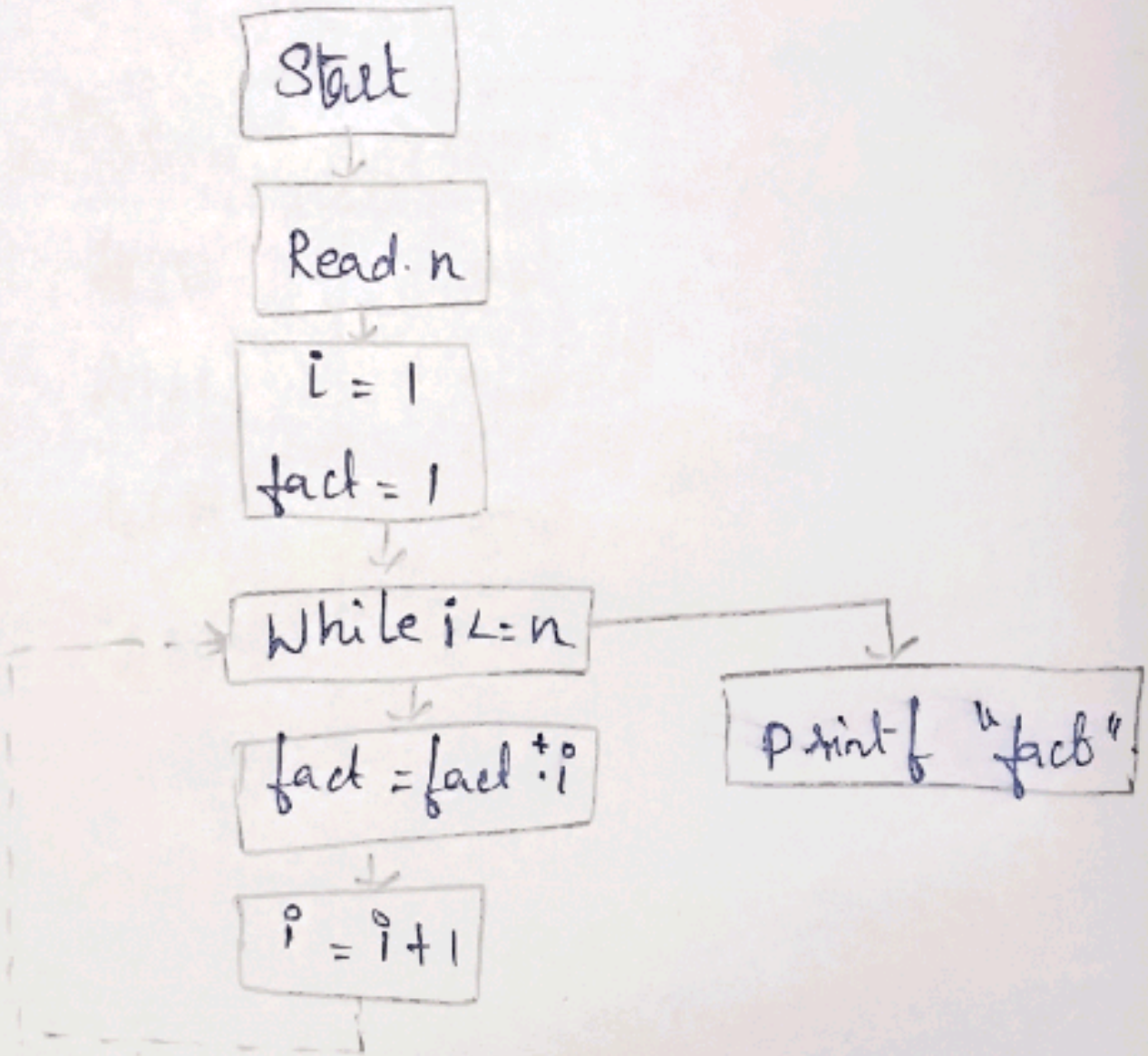
(iv) Loop : while i is less than or equal to n:

* Multiply factorial by i
* Increment i by 1

v) output : Store or print the value of factorial

vi) END.

Flowchart :



PROGRAM:

```
ORG 0000H
MOV DPTR, #4000H
MOVX A, @DPTR
MOV R0, A
INC DPTR
ACALL FACTORIAL
MOVX @DPTR, A
SJMP THIN
FACTORIAL: DEC R0
CJNE R0, #01H, PRODUCT
SJMP THICK
PRODUCT: MOV B, R0
MUL AB
ACALL FACTORIAL
THICK: RET
THIN: RET
END.
```

RESULT:

Thus, the factorial of a number was calculated and executed successfully using 8051 Keil.

Output :

40H = 18H

Manual calculation

$4! = 4 \times 3 \times 2 \times 1 = 24$

24 in Hexadecimal = 18H.